Day 2 — Pre-Deployment Planning and Sizing

Cisco AppDynamics Professional Implementer (500-430 CAPI) · 10-day course · Domain 1 (25% of the exam)

Goal: choose a Controller profile, deployment mode, Events Service layout and EUM mode from a customer scenario.
Verification: every fact in these notes was checked against the current Splunk AppDynamics on-prem documentation (October 2026). Anything not confirmed is marked **unverified**.

Blueprint objectives covered

#	Objective	Block
1.1	Plan Events Service cluster configuration and timing	5
1.2	Assess hardware needs	1, 2
1.3	Choose a Controller deployment mode	3
1.4	Choose an EUM Server deployment mode	6
1.5	Check Controller system requirements	4
1.6	Decide when to use custom correlation	7 (preview; full on Day 6)

Block 1 — How Controller sizing works

Official rule: "The best indicator of the actual workload on your Controller is the metric ingestion rate." Before installing, estimate from the **node count** (for Java, one node = one JVM). After installing, confirm with the **metric upload rate** and keep watching it as the workload grows.

Profile	Max metrics/min	Agents (approx.)	OS	CPU	RAM	Storage
Demo	10,000	5	Linux or Windows	2 cores	8 GB	50 GB
Small	50,000	50	Linux or Windows	4 cores	16 GB	400 GB
Medium	1,000,000	2,000	Linux or Windows	8 cores (16 vCPU VM)	128 GB	5 TB SSD
Large	5,000,000	15,000	**Linux only**	28 cores (32 vCPU VM)	512 GB (256 GB VM)	20 TB SSD + NVMe redo logs
Extra Large	Beyond Large	Beyond Large	—	—	—	Contact Professional Services

Exam traps: the Controller figures do **not** include the Enterprise Console or Events Service — provision those separately. A profile can only grow, never shrink. Default agent metric limits: app agent **5,000**, Machine Agent **1,450**.

Worked example: 80 licensed app agents

- Count every agent: 80 app agents + 80 Machine Agents + 1–2 Database Agents ≈ **162 agents**.
- Worst-case metrics: 80 × 5,000 = 400,000 plus 80 × 1,450 = 116,000 → ≈ **516,000 metrics/min**.
- Small fails (50 agents / 50K metrics). **Medium** fits (2,000 agents / 1M metrics). Even 80 app agents alone exceed Small.
- Headroom (rule of thumb, not official): 516K × 1.3 ≈ 670K — still within Medium.

Block 2 — Hardware and storage (1.2)

Disk **latency** matters more than raw speed, because MySQL is very sensitive to write latency.

- Primary storage write latency must stay at or below **3 ms** under sustained load. Support won't fix problems caused by slow disks.
- Avoid **NFS** (latency, risk of corruption) and **iSCSI / SAN** storage with network quality-of-service issues. Local disks usually meet requirements.
- MySQL I/O: random 16 KB InnoDB reads/writes; 128 KB log writes. Match RAID stripe size to write size, never below 16 KB.
- For ≥ 1M metrics/min on slow storage: add a mirrored NVMe write-back cache.
- Test with real-world traffic before production.

Block 3 — Controller deployment modes (1.3)

A. Where it runs: SaaS · on-premises · hybrid (on-prem Controller with AppDynamics' cloud EUM).

B. Availability: single Controller, or an HA pair (primary + secondary, active-passive). Both servers identical; EC-managed HA is Linux only; load balancer / virtual IP checks `/controller/rest/serverstatus`.

C. Tenancy:

	Single-tenant (default)	**Multi-tenant**
Accounts	One built-in account, hidden from users	Several accounts (tenants)
Login page	No account field	Extra Account field
Access	All users see all applications	Users separated by account
Use when	Most installations	Very large installs or very separate user groups
Switching	Single → Multi possible later	Multi → Single **never** possible

Install methods: EC GUI (Express or Custom), EC CLI (`platform-admin.sh`), or silent with a response file. Prep-guide sample answer: the two Linux installation modes are **silent mode and GUI installer**.

Block 4 — Controller system requirements (1.5)

- **Linux libraries:** `libaio`, `numactl` (NUMA hosts), `tzdata` (Ubuntu), ncurses 5 compatibility, `net-tools`.
- **User limits:** `nofile` 65535, `nproc` 8192 (hard and soft).
- **Antivirus:** exclude the MySQL data directory `<controller_home>/db/data`.
- **Time sync:** clocks in sync across Controller, EUM Server, Events Service nodes and monitored apps; NTP required for EUM.
- **Windows limits:** Large is Linux only; with the EC on Windows, the EC and Controller must be on the same machine.

Block 5 — Events Service: configuration and timing (1.1)

Rule	Value
Node count	Single node, or a cluster of **3 or more** (3 is the minimum for a cluster)
Machines	**Dedicated**, with identical directories, user accounts and hardware
Heap	**Half of RAM**, minimum 7 GB, **maximum 31 GB**
Disk	**SSD**, ideally 1 TB or more
Install profile	Production (default): 4 cores, 12 GB RAM, 128 GB disk · profile=dev for non-production only
Ports	9080 API (must be reachable from the EC) · 9081 admin · 9300–9400 between nodes
Retention	8, 30, 60 or 90 days — directly drives storage
Java	Bundled Java 17 from Events Service 23.2.0

Sizing by license units (official tables, AWS i2 instances): a 3-node cluster of i2.2xlarge (61 GB RAM, 8 vCPU, 1.6 TB SSD) handles about **37 Transaction Analytics** or **10 Log Analytics** license units. Log Analytics needs far more capacity per unit.

Timing

1. Plan the Events Service **before go-live**. After scaling up from the embedded instance, only **new** Database Monitoring data goes to the new cluster; old data stays unless migrated.
2. The Events Service must be running **before** EUM Analytics (Browser and Mobile Request Analytics need it).
3. Observe ingestion over **several days** — events are batched.
4. Upgrade order: Enterprise Console → **Events Service** → EUM Server → Controller.

Block 6 — EUM deployment mode (1.4)

Option	When to use it
EUM Cloud (default)	Agents send data to AppDynamics' cloud. Works with an on-prem Controller too
On-prem EUM Server — Demo mode	Demonstration and light testing only. Controller must already be installed **on the same machine**
On-prem EUM Server — Production mode	EUM and Controller on **different hosts**. Installs its **own separate MySQL**

EUM Server hardware (basic): 4 cores, 8 GB RAM (4 GB max heap), 50 GB extra disk, 10 Mbps, 64-bit Linux or Windows. Handles about 10K page requests/min or 10K concurrent mobile users. On-prem Analytics raises requirements, especially disk. NTP on both EUM and Controller.

Peak beacons per minute (beacon every 10 s)	RAM
~3K	8 GB
~16K–26K	16 GB
~33K and above	32 GB

Block 7 — Custom correlation (1.6), preview

Agents keep a transaction together across tiers by adding a correlation header to **supported** protocols. Use **custom correlation** when a call goes over an **unsupported protocol or framework** (a custom socket, a proprietary messaging library). Without it, the downstream tier appears as a separate BT. Prep-guide sample answer: needed for calls that **use unsupported frameworks**. Full detail on Day 6.

Sizing exercise — worked answers

Scenario	Answer	Why
300 agents, 400K metrics/min, Windows only	**Medium**	Small fails both limits; Medium allows 2,000 agents / 1M metrics and supports Windows
4,000 agents, 3M metrics/min	**Large, on Linux**	Medium tops out at 2,000 agents / 1M metrics; Large is Linux only
Analytics, 90-day retention, Large Controller	**Separate dedicated cluster, ≥ 3 nodes, SSD**	Never on the Controller host (prep-guide answer: a separate cluster)

Lab (~40 min)

1. **Transaction thresholds:** In your SaaS Controller: application → Configuration → Slow Transaction Thresholds. Record the default values for slow, very slow and stall.
2. **Tenancy:** Look at your SaaS login URL and page — is there an account name? (SaaS is multi-tenant.)
3. **Paper design:** A bank has 120 Java JVMs, 120 Machine Agents and 2 Oracle databases, wants Transaction Analytics for 30 days and Browser RUM at 8K page requests/min. Write down: Controller profile and OS, HA or not, Events Service layout, EUM mode and RAM.

Day 2 Quiz — with answer key

Cover the answer lines and test yourself first.

Q1. What is the primary consideration when sizing an on-prem Controller?

A. Number of business applications
B. Number of Controller users
C. Number of health rules
D. Metric ingestion rate (metrics per minute)
Answer: D — The docs call the metric ingestion rate the best indicator of Controller workload.

Q2. 1,500 agents, 800,000 metrics/min, Controller must run on Windows. Which profile?

A. Small
B. Medium
C. Large
D. None, because Windows is not supported
Answer: B — Medium supports ~2,000 agents and 1M metrics/min on Linux or Windows. Large is Linux only.

Q3. Which Controller profile is supported on Linux only?

A. Demo
B. Small
C. Medium
D. Large

Answer: D — Demo, Small and Medium support Linux or Windows; Large is Linux only.

Q4. Maximum sustained write latency allowed for Controller primary storage?

A. 1 ms
B. 3 ms
C. 10 ms
D. 20 ms

Answer: B — At or below 3 ms under sustained load. Avoid NFS and SAN/iSCSI with QoS issues.

Q5. A Controller was installed multi-tenant; the customer wants single-tenant. What can they do?

A. Switch in the Administration Console
B. Switch with platform-admin.sh
C. Nothing — multi-tenant to single-tenant is not possible
D. Switch after upgrading the Controller

Answer: C — Single → multi is possible later; multi → single never is.

Q6. Events Service heap for a node with 128 GB RAM?

A. 64 GB (half of RAM)
B. 31 GB
C. 7 GB
D. 96 GB

Answer: B — Half of RAM, minimum 7 GB, maximum 31 GB — the cap applies.

Q7. Which statement about Events Service clusters is correct?

A. Minimum of 2 nodes
B. Minimum of 3 nodes, on dedicated machines with identical hardware
C. Nodes should share the Controller host
D. Nodes can differ in hardware if total RAM is enough

Answer: B — Single node or 3+ nodes, dedicated machines with identical directories, accounts and hardware.

Q8. After scaling up from the embedded Events Service, what happens to existing Database Monitoring data?

A. It moves automatically
B. It is deleted
C. It stays in the embedded instance unless migrated
D. It is copied to the Controller's MySQL

Answer: C — Only new data goes to the new cluster — plan the cluster before go-live.

Q9. When must you choose Production mode for the on-prem EUM Server?

A. When EUM Server and Controller are on different hosts
B. When using Browser RUM only
C. When fewer than 3K beacons/min are expected
D. When the Controller is single-tenant

Answer: A — Demo mode needs the Controller on the same machine; Production uses its own host and separate MySQL.

Q10. Custom correlation is required to keep transaction context for calls that:

A. Use HTTP between two Java tiers
B. Use unsupported protocols or frameworks
C. Go to a database backend
D. Run inside the same JVM

Answer: B — Supported protocols correlate automatically; unsupported ones need custom correlation.

Quick-review cheat sheet

Remember	Value
Primary sizing driver	Metric ingestion rate; second: number of agents
Profiles (metrics/min · agents)	Demo 10K·5 · Small 50K·50 · Medium 1M·2,000 · Large 5M·15,000
Linux only	Large profile; EC-managed HA
Agent metric limits	App agent 5,000 · Machine Agent 1,450
Disk	Write latency ≤ 3 ms · no NFS · avoid iSCSI/SAN with QoS issues
Tenancy	Single-tenant default · single → multi OK · multi → single never
Events Service	1 node or ≥ 3 · dedicated identical machines · heap ½ RAM, 7–31 GB · SSD ≥ 1 TB
Events Service ports	9080 API · 9081 admin · 9300–9400 cluster
Retention options	8 / 30 / 60 / 90 days
EUM modes	Cloud (default) · Demo (same host as Controller) · Production (own host, own MySQL)
EUM basic hardware	4 cores · 8 GB RAM (4 GB heap) · 50 GB · ~10K page requests/min
Custom correlation	Unsupported protocols or frameworks

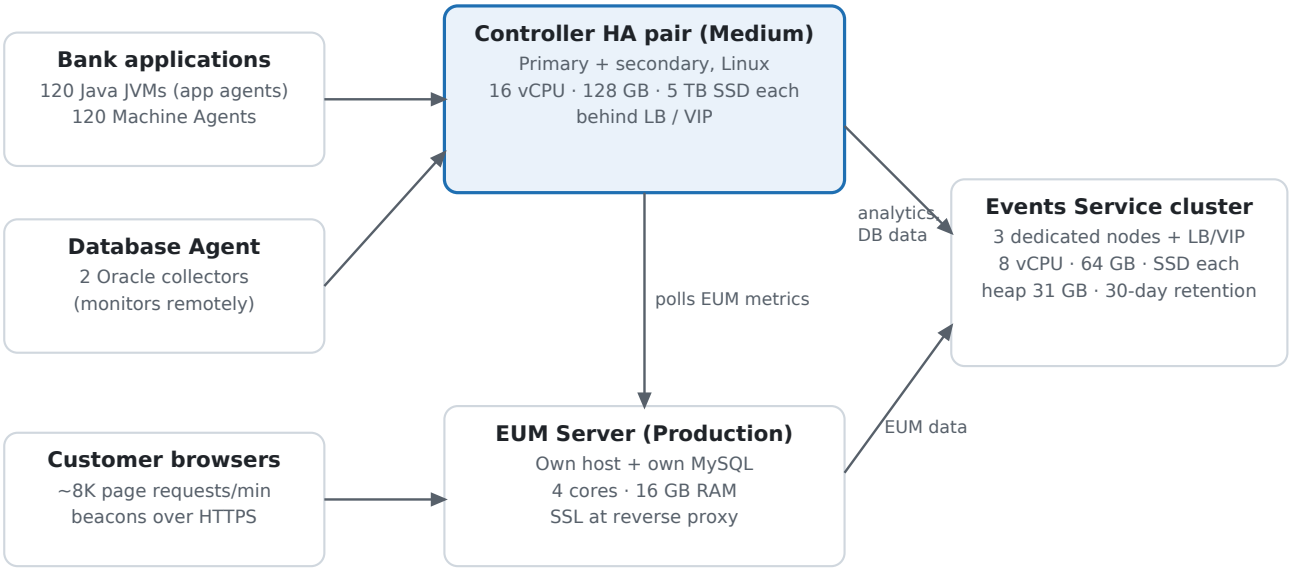
Sources

- Controller Sizing (26.9.0)
- Disk I/O Requirements
- About Controller Tenancy Mode
- Events Service General Requirements
- Events Service Node Sizing by License Units
- EUM Server Hardware Requirements
- Metrics Limits
- All Other Traffic Business Transactions

Model answer — bank deployment design (Day 2 lab task 3)

Scenario: 120 Java JVMs, 120 Machine Agents, 2 Oracle databases; Transaction Analytics with 30-day retention; Browser RUM at about 8K page requests per minute. This is how an implementer would answer it, step by step, with the rule behind each choice.

Bank design: HA Controller pair, dedicated Events Service cluster, production EUM Server



1. Controller profile and OS → Medium, on Linux

Step	Working
Count agents	120 app agents + 120 Machine Agents + 1 Database Agent (2 Oracle collectors) ≈ **241 agents**
Worst-case metrics	120 × 5,000 = 600,000 · 120 × 1,450 = 174,000 → **≈ 774,000 metrics/min** (default agent limits; the Database Agent adds a small amount)
Pick profile	Small fails (50 agents / 50K metrics). **Medium** fits (2,000 agents / 1,000,000 metrics/min)
Headroom	774K × 1.3 ≈ 1.0M — right at Medium's limit (rule of thumb, not official). Real agents usually register fewer metrics than their limit, so run a pilot and confirm with the metric upload rate. If the sustained rate approaches 1M, move to **Large**
OS	**Linux.** Medium allows Windows, but Linux is required for Enterprise Console-managed HA and keeps the path to Large (Linux only) open
Hardware (each server)	8 cores bare metal / 16 vCPU VM · 128 GB RAM · 5 TB SSD · write latency ≤ 3 ms · no NFS

2. High availability → Yes, an HA pair

- A bank's monitoring platform is business-critical, so a single Controller is a single point of failure.
- **Primary + secondary** Controller, active-passive, with **identical** OS, CPU, RAM and disk (two Medium servers).
- Linux only for EC-managed HA; passwordless SSH between hosts; both hostnames in each /etc/hosts; same database root password.
- Agents and users connect through a **load balancer / virtual IP** that checks `/controller/rest/serverstatus` and sends traffic to one Controller only (no round-robin).

- Run `activate-ha-modules` so failover does not depend on the Enterprise Console.

3. Events Service layout → dedicated 3-node cluster

- **Dedicated cluster of 3 nodes** (the minimum for a cluster), never on the Controller hosts, with a load balancer / VIP in front.
- Node size, using the official license-unit table as the reference: i2.2xlarge-class nodes (8 vCPU, ~61–64 GB RAM, ≥ 1.6 TB SSD). A 3-node cluster of this size handles about **37 Transaction Analytics license units**.
- **Heap:** half of 64 GB = 32 GB, so the **31 GB maximum** applies.
- **Retention:** 30 days (allowed values are 8, 30, 60 or 90). Storage grows with retention.
- If the bank's Transaction Analytics license exceeds the table's figure, scale out (5 nodes ≈ 44 units) or use larger nodes (3 × i2.4xlarge ≈ 41 units). Confirm the purchased license-unit count — it drives this choice.
- The cluster also stores Database Monitoring data from the 2 Oracle databases, and Browser Request Analytics if enabled — include that load.
- **Timing:** build it before go-live (scaling up later leaves old Database Monitoring data behind) and observe ingestion over several days.

4. EUM mode and RAM → Production mode, 16 GB RAM

- **Production mode** on its **own host** (installs its own MySQL). Demo mode is ruled out: it is for light testing and must share the Controller's machine.
- Base hardware (4 cores, 8 GB) handles about **10K page requests/min**. 8K/min is about 80% of that, so a bank should not run at the minimum: use **16 GB RAM** (the next tier in the beacon table, ~16K–26K beacons/min) with 4 cores.
- Disk: 50 GB base, plus room for resource-timing snapshots (default up to 1K/min kept 15 days → about 64 GB or more recommended).
- Browsers on the internet send beacons over HTTPS: terminate SSL at a **reverse proxy** in front of the EUM Server.
- NTP on both the EUM Server and the Controller. Point the Controller's eum.* properties at the EUM host (Day 5).

Exam-style summary answer

Controller: Medium profile on Linux · **HA:** yes, active-passive pair of identical Linux servers behind a load balancer · **Events Service:** separate dedicated 3-node SSD cluster, heap 31 GB, 30-day retention · **EUM:** on-prem Production mode on its own host, 16 GB RAM (8 GB minimum).

> **Judgement calls (not fixed rules):** the 30% headroom, choosing 16 GB over the 8 GB minimum for EUM, and the Events Service node size, which depends on the purchased license units. Everything else follows a documented rule from this session.